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Comparison of Infrasound Wind Noise Reduction Systems for Use in Temporary Deployments

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ABSTRACT

Infrasound, or low frequency sound ≤ 20 Hz, is produced by a variety of natural and anthropogenic sources. Wind also generates signals within this frequency band and serves as a persistent source of infrasonic noise. Infrasound sensors measure pressure fluctuations, which scale with the ambient density and velocity fluctuations of ground winds. Here we compare four different wind noise reduction systems, or "filters", and make recommendations for their use in temporary infrasound deployments. Our results show that there are two filters that are especially effective at reducing wind noise: (1) a Hyperion high frequency (HF) shroud with a 1 m diameter metal mesh dome placed on top and (2) a Hyperion Four Port Garden Hose shroud with 4 Miracle-Gro Soaker System garden hoses. We also find that placing a 5-gallon bucket over the HF wind shroud should not be done as it provides a negligible decrease in noise up to ~ 1 Hz and then an increase in noise. We conclude that it is up to the researcher to determine which of the other filters is best for their needs based on location and expense. We anticipate this study will be used as a resource for future deployments when a wind noise reduction method is necessary but only needed for a limited time period.

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ACRONYMS AND DEFINITIONS

Abbreviation	Definition
CTBTO PrepCom	Comprehensive Nuclear-Test-Ban Treaty Organization
IMS	International Monitoring System
HF	High Frequency
PSD	Power Spectral Density
SNR	Signal-to-Noise Ratio

1. INTRODUCTION

Infrasound is low frequency sound ≤ 20 Hz. It is produced by a variety of natural and anthropogenic sources including volcanoes, earthquakes, chemical explosions, and rockets. Monitoring these sources is important for national security and hazard mitigation and often requires recording high fidelity waveforms. However, the state of the atmosphere can make this difficult. As described by Raspet et al. (2019), "the greatest impediment to the detection and analysis of infrasound signals are the intrinsic pressure fluctuations due to air turbulence, known as wind noise, always present in the atmosphere". For the low frequencies of the infrasound band, this type of noise can easily dominate the signal. Two types of wind noise exist: stagnation pressures and intrinsic pressures. Stagnation pressures are generated via turbulence acting with an object (i.e. the infrasound sensor). Intrinsic pressures can further be grouped into turbulence-turbulence interaction pressures (turbulence interacting within the atmosphere) and turbulence-shear interaction pressures (turbulence interacting with wind shear near the ground) (Raspet et al., 2019). Wind noise reduction systems aim to decrease both of these types of wind noise.

The utility of wind noise reduction systems, or "filters", for infrasound has been studied extensively. The first filter consisted of a 602 m long tapered pipe with 100 openings that reduced noise below 1 Hz, but the resulting directionality of this system was not well suited for characterizing unknown signal backazimuths (Daniels, 1959). Since then, there have been many efforts to develop wind filters for the International Monitoring System (IMS) infrasound stations, which are used by the Preparatory Commission for the Comprehensive Nuclear-Test-Ban Treaty Organization (CTBTO PrepCom) to monitor for nuclear explosions. Many of the infrasound wind filters on the IMS infrasound stations are metal rosette pipe arrays, which work by averaging the signal over a large area (Hedlin and Raspet, 2003; Raspet et al., 2019). Pipe arrays allow for a reduction of wind noise up to 20 dB (Raspet et al., 2019). However, they are high cost, difficult to deploy, and occupy a large footprint, making them unsuitable for temporary or fast infrasound deployments. Porous hoses are often used as an inexpensive and light alternative, despite Walker & Hedlin (2010) recommending against their use over ten years ago because the hoses are fragile and their response is unknown. Recorded infrasound signals using a porous filter have been shown to be either attenuated or amplified depending on the signal backazimuth, incidence angle, and frequency content (Howard et al., 2007). A second type of filter, porous wind domes, have shown promise in reducing wind noise by up to 25 dB (Noble et al., 2014; Raspet et al. 2019). In particular, porous wind domes are recommended by Raspet et al. (2019) as they provide wind noise reduction along with better environmental protection, though reducing wind noise at lower frequencies requires a large footprint (Raspet et al., 2019). A third wind noise reduction method that is occasionally used consists of placing a bucket over the infrasound sensor, though only anecdotal evidence exists as to how this method compares to the others. There are no professional recommendations for placing a bucket over the infrasound sensor, but the method has caught on in some circles and it is often not noted in publications. In fact, the authors have both used this method and have noticed its use from images in conference presentations. Focusing on the frequency band of interest for a variety of infrasound monitoring applications, 0.01-20 Hz, we (1) compare the wind noise reduction ability of the aforementioned filters and (2) make recommendations regarding which method to use in temporary infrasound deployments in a variety of field settings.

2. EXPERIMENT SETUP

The experiment consisted of five co-located Hyperion Technology Group (HTG) IFS-3000 infrasound sensors connected to a Trimble Reftek RT-130 digitizer. Data was sampled at 100 Hz. Five sensors were equipped with wind filters: (1) a simple Hyperion High Frequency (HF) shroud, which will be referred to as the "bare sensor" as it does little to reduce wind noise, (2) a second sensor with an HF shroud which produced similar results and therefore will not be further discussed, (3) a Hyperion HF shroud with a 5-gallon bucket placed on top and a hole drilled in the side for the cabling, (4) a Hyperion HF shroud with a 1 m diameter metal mesh dome placed on top, and (5) a Hyperion Four Port Garden Hose shroud with four Miracle-Gro Soaker System garden hoses with push on fittings. The latter filter is about 30 m in diameter and will be referred to as the "porous hose filter". The network layout is shown in Figure 2-1. Note that the porous hose filter system also makes use of a 5-gallon bucket. However, in this case the bucket serves as a weather barrier and wind noise reduction is done solely by the porous hoses. In situ wind speed data was also collected at the site. The sample rate of this data was irregular so it was resampled to 0.0008 Hz (20 min/sample) using linear interpolation. All sensors recorded data from January 1 through April 22, 2019.

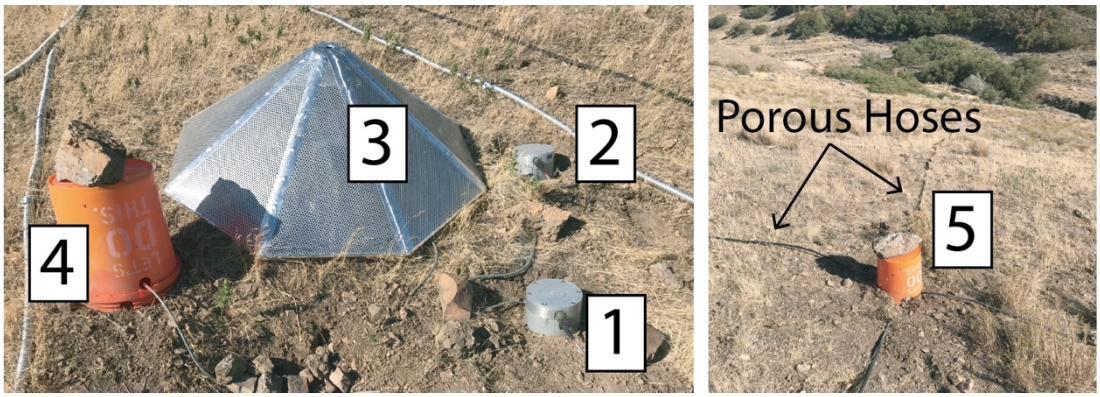


Figure 2-1. The wind noise reduction system test configuration. The systems shown are (1) a Hyperion HF shroud, known as the "bare sensor", (2) a second sensor with a Hyperion HF shroud (not discussed due to similarity of results), (3) a Hyperion HF shroud with a 1 m diameter metal mesh dome placed on top, (4) a Hyperion HF shroud with a 5-gallon bucket placed on top and a hole drilled in the side for cabling, and (5) a Hyperion Four Port Garden Hose shroud with 4 Miracle-Gro Soaker System garden hoses with push on fittings, known as the "porous hose filter". The porous hose filter was placed as close as possible to the other group of sensors. In situ wind speed data was also collected at the site though this picture was taken before that sensor was deployed.

3. METHODS

Data was first divided into wind speed intervals. Following the method of Hedlin & Walker (2003), wind speed intervals were broken into 0-0.5 m/s, 1-1.5 m/s, 2-2.5 m/s, 3-3.5 m/s, 4-4.5 m/s, and 5-5.5 m/s. For each wind speed interval, power spectral densities (PSDs) were calculated using a 20 minute Hann window with 50% overlap. The average PSD was then taken for each wind speed interval for each wind filter. PSDs were converted to decibels using the transformation $10\log_{10} x$ where x corresponds to the calculated PSD values. The mean PSD was found for each wind filter and wind speed interval. Then the reduction in overall noise for each filter was calculated by the subtracting each filter's PSD from the bare sensor PSD.

4. RESULTS

We analyzed the noise reduction for each filter for the six wind speed intervals. The porous hose filter provides the largest decrease across the entire frequency band of interest. The overall decrease in noise power ranges 2-15 dB between 0.01-20 Hz and is consistent with the findings discussed in Raspet et al. (2019). The metal wind dome reduces noise as well, with a range of 2-7 dB between 0.01-20 Hz. Surprisingly, the bucket system shows no reduction in noise power between 0.01-1 Hz and actually shows an increase in noise above 1-3 Hz, depending on the wind speed interval. Figure 4-1 shows mean noise reduction for all filter systems and all wind speed intervals.

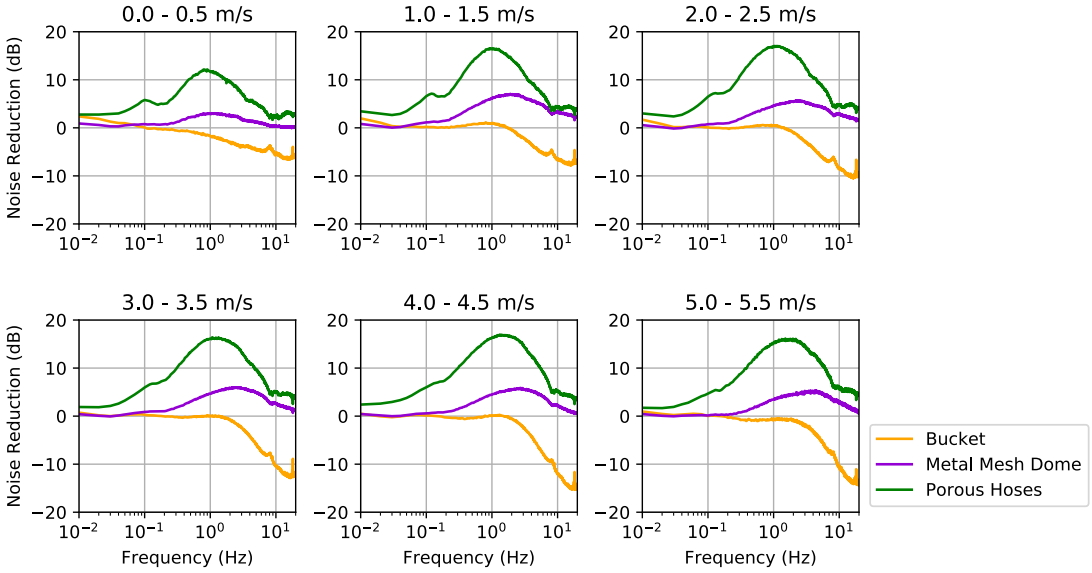


Figure 4-1. Difference in mean power spectral densities. The porous hose filter provides the greatest overall decrease in noise between 0.01-20 Hz and is consistent with the findings discussed in Raspet et al. (2019). The metal wind dome reduces noise as well. The bucket system shows an increase in noise above 1-3 Hz, depending on the wind speed interval.

5. DISCUSSION

As described in the previous section, the porous hose filter is the most effective at decreasing overall noise. However, Howard et al. (2007) noted a change in amplitude and frequency content of the signal recorded using this type of filter that may be dependent on signal backazimuth or other factors. This has important implications for explosion monitoring at regional to global distances as a weak signal could be further attenuated by the system. Differences in amplitude and frequency content will also translate into errors for any research that requires waveform fidelity such as yield calculation, source characterization, waveform inversion, and beamforming. This issue should be explored more, and a transfer function computed for the porous hose filter that would allow for analysis of the response due to backazimuth, incidence angle, and frequency content. When waveform fidelity is of importance, a transfer function would also allow for the recovery of historical data recorded at sites using the porous hose filter.

Due to the aforementioned findings of the porous hose filter, we recommend using the metal mesh dome for temporary wind noise reduction systems. The main benefit of the system is that its footprint is only 1 m. It can be placed in areas where space is limited such as on a ledge or in a populated area. It is also possible that the addition of gravel around the base of the metal mesh dome may improve its noise reduction capability as wind may be entering at the base. However, it is important to note that the metal mesh dome weighs about 50 lbs and requires at least two people to transport. This would make it particularly difficult to carry up a steep slope. Fabric domes may be better suited for these types of deployments as they are lighter and have shown a similar reduction in noise by Raspet et al. (2019), though an analysis of these filters is outside the scope of this study.

Results from analysis of the bucket filter suggest that it should never be used. It is unlikely that the increase in overall noise above 3 Hz is due to wind deforming the bucket since this effect happens at the lowest wind speed interval, suggesting it is intrinsic to the system. With the bucket, we are effectively sampling only one point in space (the hole for the sensor cable), rather than averaging over multiple points in space (as is done by 16 holes surrounding the circumference of the HF shroud on the bare sensor). Placing the bucket, with only one hole drilled in the side, is effectively canceling the inherent noise reduction capability of the bare sensor. Drilling many holes into the bucket could provide wind noise reduction capability, but it is unclear whether it would outperform the metal mesh dome or porous hose filters.

While the bare sensor is outperformed by the metal mesh dome and porous hose filters, it is important to point out that it has the smallest footprint and is the lightest out of all of the systems. If signals of interest are expected to have a high signal-to-noise ratio (SNR) then the wind noise reduction system becomes less important. For example, local infrasound monitoring of volcanic eruptions would likely produce high SNR recordings. Therefore, placing a bare sensor in vegetation would allow the researcher to collect the signals of interest without the use of a wind filter. Lastly, these wind filters are commonly used in infrasound deployments, but are not the only option. Other wind noise reduction systems, such as fabric wind domes, require an extensive study to provide recommendations on their use.

Degradation of the porous hose filter likely occurs over time due to environmental impacts such as rain, ultraviolet radiation, and destruction by domesticated animals and/or wildlife. Continuous data recording during a longer duration of time would allow for a more robust analysis, possibly providing a better understanding of how each of the wind noise reduction filters perform over time. In such a case, it might be possible to identify a time when the frequency response suffers serious

degradation. It is also recommended that the frequency response of reference sensor be measured both before and after deployment to quantify any changes.

6. CONCLUSIONS

The porous hose filter decreases overall noise, but suffers from an unknown frequency response that may alter waveform amplitude and frequency content. The metal mesh dome both decreases overall noise and preserves waveform fidelity, and thus is the recommended wind noise reduction system for future experiments. However, the dome is heavy. A fabric dome would likely perform similarly well and be more transportable. Placing a bucket over the bare sensor does nothing to decrease noise and actually increases noise above 1-3 Hz, so it is strongly recommended against. Of course, if high SNR is expected then the wind filter becomes less important and the bare sensor can be used on its own. We urge future studies to (1) use Table 1-1 to make decisions regarding a wind filter for future experiments and (2) generate a transfer function for the porous hose filter to analyze its response as a function of backazimuth, incidence angle, and frequency content.

Table 6-1. Wind Noise Reduction System Recommendations

Station Characteristics	Recommended Wind Noise Reduction System
Low-lying vegetation, ≥ 30 m space	• If waveform fidelity not an issue: Hyperion Four Port Garden Hose shroud + 4 Miracle-Gro Soaker System hoses • Otherwise: Hyperion HF shroud + metal mesh dome
Low-lying vegetation, ≤ 30 m space	• Hyperion HF shroud + metal mesh dome
Dense vegetation, ≥ 30 m space	• If waveform fidelity not an issue: Hyperion Four Port Garden Hose shroud + 4 Miracle-Gro Soaker System hoses • Otherwise: Hyperion HF shroud + metal mesh dome
Dense vegetation, ≤ 30 m space	• Hyperion HF shroud + metal mesh dome
On top of a steep slope	• If waveform fidelity not an issue and space is available: Hyperion Four Port Garden Hose shroud + 4 Miracle-Gro Soaker System hoses • Otherwise: Hyperion HF shroud + metal mesh dome (if able to carry) OR fabric mesh dome
Expecting high SNR signals	• Hyperion HF shroud

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